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Maven: (Build tool)

[Ant - build
gradel - net]

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Introduction:

1. Maven is an automation and management tool
2. It is written in java language and used to build and manage projects written in any other languages.

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③ Maven helps the developer to create a java-based project more easily

④ To configure maven, we need to use Project Object Model, which is stored in pom.xml file.

What maven does:

① We can create our builds for any environment

② Downloading dependency and also pulling other components.

③ Work with local repository i.e, c:\users\Yashaswi\.m2

④ Work with IDE (Integrated development environment)

⑤ It can work with CI tools like jenkings.

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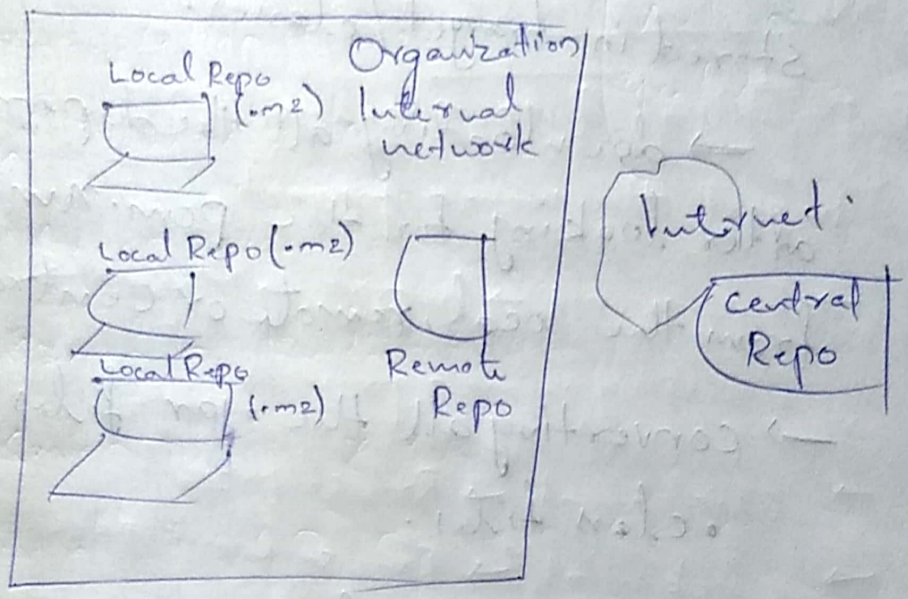
Maven Architecture:



Maven build life cycle:

① **Default (build)**: Used to create the application

② **clean**: cleans up the artifacts that are created by previous builds

③ **Site**: Generates site documentation for the project

Maven Phases or goals:

① **clean**: Delete the target directory

② **validate**: Validate, if the project is correct.

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③ Compile: Compile source code, classes stored in target/classes.

→ downloading all the dependencies after looking for the pom.xml file from the local, remote or central repo

→ converting all the .jar files to .class files.

④ Test: run tests (eg. unit testing)

⑤ Package: take the compiled code and package it in its distributable format

eg. JAR, WAR.

⑥ Verify: run any checks to verify the package is valid and meets quality criteria.

⑦ Install: install the package into the local repository (.m2)

→ moving the WAR files to .m2

⑧ Deploy: copies the final package to the remote repository (using jenkins)

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Install maven in windows environment:

(latest)

① Download JDK, maven and extract it

② Add JAVA_HOME and MAVEN_HOME in environment variable

③ Add java and maven path in environment variable

④ Verify java and maven.

After download & installing:

Steps: MyPC → properties → Advanced System Settings

→ Environment Variables → System Variables.

→ Add new (JAVA_HOME - path
MAVEN_HOME - path)

Reference: mkyong.com

* By setting the env variables, Java & maven will work throughout the environment. Unlike only where it is installed.

→ In cmd prompt (as administrator) or in gitbash

→ java -version
→ mvn -version

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Maven webapp in eclipse in windows environment :

① Download & open eclipse IDE

② Import the maven project

→ File menu → Import → Maven → Existing maven projects.

→ click next → Browse project

→ click next → Browse workspace

→ click on finish.

→ Window → preferences → add m2 → path

③ Run the maven web project

→ Right click on project → Run As

→ Run on Server.

→ Adding JDK.

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Creating sample maven project using eclipse

① Download elipse & open it

② File menu → New → Project → Maven → Maven project → Next → Next.

③ Now write the group ID, artifact ID, Package as shown in next slide.

④ click on finish.

→ Archetype → Giving the details of the pom file in the beginning of the project.

⑤ All the files will be created automatically such as HelloJava files pom.xml file, test case file etc...

⑥ The directory structure of the maven project is shown in the below figure:

⑦ Now we can see the code of App.java file and run it

⑧ If we right click on the project → Run as, we will see the maven options to build the project.

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Build source code by using maven in Linux environment:

1. Launch the ubuntu 16 machine from AWS
2. Update the machine
3. Install Java and check
4. Install git and check
5. Install maven and check.
6. Login to Github ~~and~~ account and copy the url of the repository
7. Clone the repository using git terminal
8. Execute the maven goals.

Steps:

1. Connect to the machine as root user

   → sudo -i

2. Update: → apt-get update
3. Install java (digital ocean - google search)

   → apt-get install default-jdk -y

   → java -version (checking)

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④ Install maven

→ apt-get install maven -y

→ mvn -version (checking)

⑤ From Ubuntu 16 version onwards, git is installed by default.

→ In putty: apt-get install git -y.

→ git --version (checking version)

⑥ → git clone <repo url>

→ cloning the repo from github

→ ls

→ cd <directory name>

→ ls

→ make sure that pom.xml & all the required files are present.

⑦ Maven goals.

① → mvn compile.

→ Looks for the pom file, downloads the dependencies & converting all the .jar files to .class files in target folder.

② → mvn test

performs any tests

③ → mvn package.

Building war file.

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The .war file is generated in target folder.

(4) → mvn install

The war file is moved to the .m2 folder from target folder.

(5) To see hidden folders in linux

→ ls -la (being in root)

(6) cd .m2/

ls

cd ..... We find the .war file

→ The stable .war files are moved to the artifactory.

→ After the approvals, .war files are moved to the production server using ansible.

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